



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to Sri Valli Agencies

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

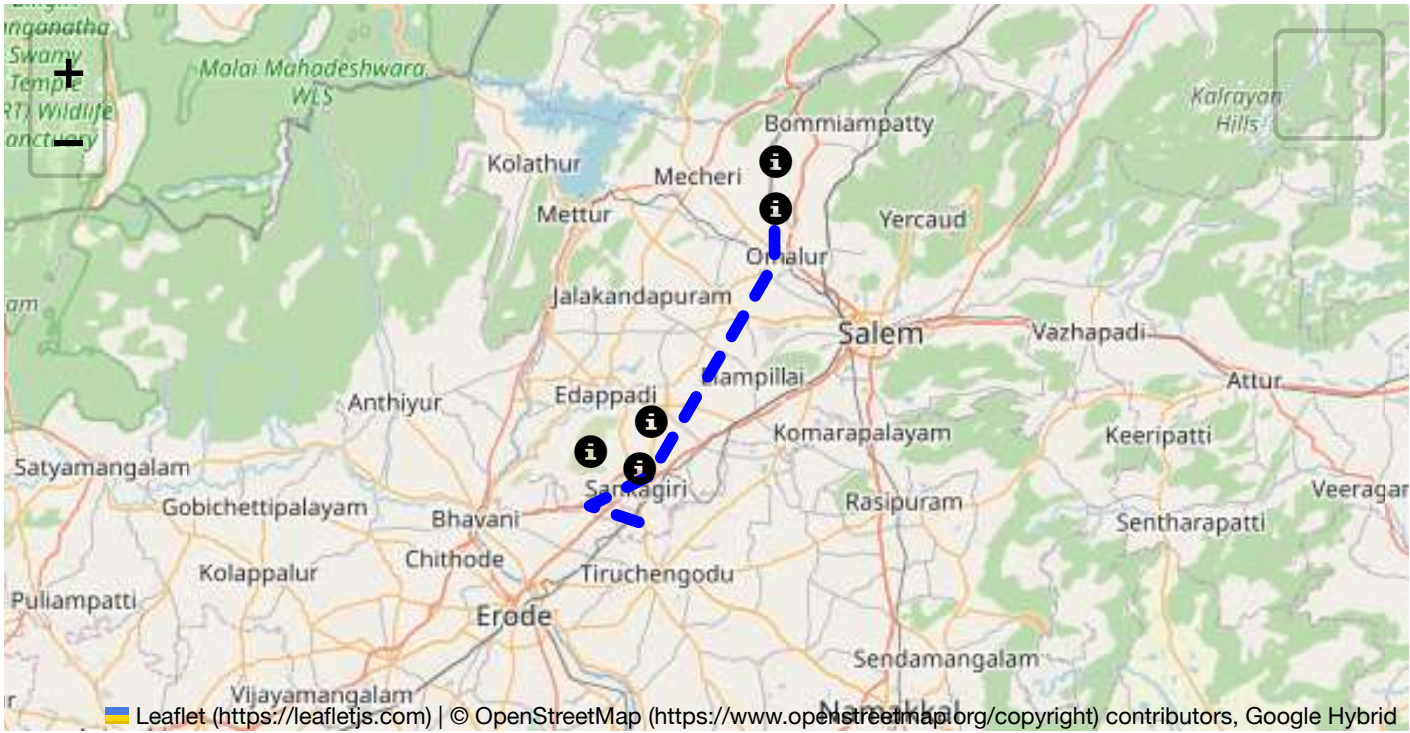
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

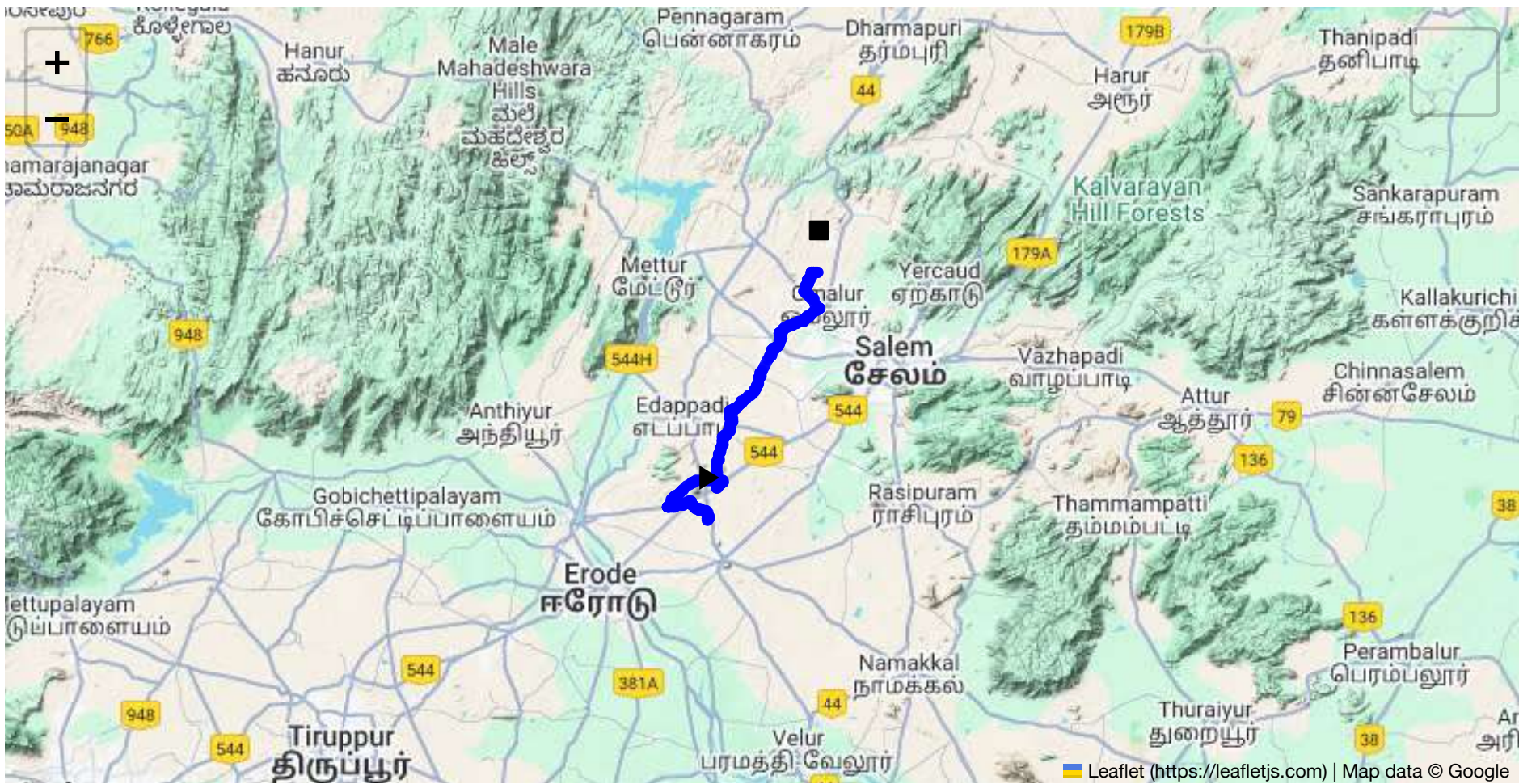
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 64.43 km
Estimated Duration: 1.6 hours
Adjusted Duration (Heavy Vehicle): 2.0 hours

Start: (11.4381, 77.8734)
End: (11.789657, 78.032799)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from CVQF+23W, Sangagiri to 235/1C, Semmandapatti passes through the following coordinate check-points:

- FR48+7XQ, Bhavani Main Road, Padaiveedu
- FVRQ+J4R, Aiveli
- P2PM+M9M, Salem

This route spans approximately 64.43 kilometers and is primarily a combination of major highways and arterial roads, commonly used for both local traffic and transportation of goods.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu experiences a hot tropical climate with three main seasons:

- Summer (March to June):** High temperatures can affect road conditions, causing tar to soften and increasing the risk of tire blowouts.
- Monsoon (June to September):** Heavy rains can lead to waterlogging, reduced visibility, and slippery roads.
- Winter (November to February):** Mild and dry, posing fewer weather-related risks.

Weather patterns suggest the monsoon season introduces the highest hazard potential due to reduced traction and possible temporary flooding.

3. Analysis of Traffic Patterns

- **Peak Hours:** Typically occur in the mornings (8:00-10:00 AM) and evenings (5:00-7:00 PM).
- **Congestion-Prone Areas:**
 - Bhavani Main Road, especially near Padaiveedu, tends to have slow-moving traffic.
 - Entry and exit points in Salem city experience congestion, especially during peak hours.

4. Assessment of Road Quality and Infrastructure

- **Highways and Main Roads:** Generally maintained with occasional potholes post-monsoon.
- **Urban Sections:** More prone to road damage due to heavy vehicular activity.

5. Suggestions for Alternative Routes for Emergencies

- **State Highway 79:** Provides an alternate northbound route bypassing congested areas.
- **Local Bypasses:** Utilize the network of rural roads around Salem for small segments in emergencies.

6. Summary of Local Regulations Affecting Hazardous Material Transport

- Transport of hazardous materials is regulated under strict time windows (often restricted to nighttime in populated urban areas).
- Special permits are required, and mandatory safety checks are enforced.

7. Overview of Historical Incidents

- There have been incidents involving overturns and spills, often linked to speeding or adverse road conditions during the monsoon season.

8. Environmental Considerations and Sensitive Areas

- Passing through agricultural regions; precaution is needed to prevent contamination of farmlands.
- Proximity to residential areas in Salem may increase the risk of impact in incidents.

9. Analysis of Communication Coverage

- Communication networks are generally reliable in urban areas.
- Potential dead zones may exist in rural segments between Aiveli and Padaiveedu.

10. Estimated Emergency Response Times

- **Urban Areas (Salem):** Approximately 15-20 minutes.
- **Rural Stretch:** May extend to 30-45 minutes due to distance from major centers.

11. Overall Summary of Risk Assessment

- **Low to Moderate Risk on Highway Sections:** Good road conditions and communication coverage.
- **Moderate to High Risk in Monsoon Seasons:** Due to weather-related hazards, particularly waterlogging and poor traction.
- **High Risk in Urban Areas During Peak Hours:** Increased accident potential due to congestion.

- **Need for Continuous Weather Monitoring:** Planning requires adaptability to weather conditions, especially during the monsoon season.

Overall, careful planning, adherence to local regulations, and ongoing situational awareness are crucial to mitigate potential risks along this route. Proper training for truck drivers focused on handling emergencies and understanding traffic patterns would be essential in ensuring safety.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	Medium	11.43822, 77.87348	30 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44029, 77.87544	15 KM/Hr
3	Turn	Medium	11.44900, 77.87406	30 KM/Hr
4	Turn	Medium	11.45358, 77.85781	30 KM/Hr
5	Turn	High	11.45353, 77.85697	15 KM/Hr
6	Turn	Medium	11.45853, 77.85139	30 KM/Hr
7	Turn	Medium	11.46307, 77.84941	30 KM/Hr
8	Turn	Medium	11.46316, 77.84921	30 KM/Hr
9	Turn	High	11.45542, 77.81725	15 KM/Hr
10	Blind Spot	Blind Spot	11.45631, 77.81668	10 KM/Hr
11	Turn	High	11.49224, 77.89606	15 KM/Hr
12	Turn	High	11.49153, 77.89615	15 KM/Hr
13	Blind Spot	Blind Spot	11.48364, 77.88780	10 KM/Hr
14	Turn	Medium	11.49155, 77.88776	30 KM/Hr
15	Turn	Medium	11.49441, 77.88495	30 KM/Hr
16	Turn	Medium	11.49469, 77.88488	30 KM/Hr
17	Turn	High	11.55614, 77.89764	15 KM/Hr
18	Turn	Medium	11.58218, 77.90845	30 KM/Hr
19	Turn	Medium	11.58285, 77.90790	30 KM/Hr
20	Turn	High	11.58317, 77.90647	15 KM/Hr
21	Turn	Medium	11.63840, 77.94820	30 KM/Hr
22	Turn	Medium	11.64217, 77.95200	30 KM/Hr
23	Turn	Medium	11.67858, 77.96834	30 KM/Hr
24	Turn	Medium	11.68925, 77.97742	30 KM/Hr
25	Turn	High	11.68927, 77.97748	15 KM/Hr
26	Turn	Medium	11.70218, 77.97866	30 KM/Hr
27	Turn	High	11.70227, 77.97864	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
28	Turn	Medium	11.72565, 78.01770	30 KM/Hr
29	Blind Spot	Blind Spot	11.73785, 78.03551	10 KM/Hr
30	Turn	High	11.76193, 78.01473	15 KM/Hr
31	Turn	Medium	11.77342, 78.01842	30 KM/Hr
32	Turn	High	11.77367, 78.01819	15 KM/Hr
33	Turn	Medium	11.77764, 78.01932	30 KM/Hr
34	Turn	Medium	11.77800, 78.01975	30 KM/Hr
35	Turn	Medium	11.77858, 78.02002	30 KM/Hr
36	Turn	Medium	11.77915, 78.02111	30 KM/Hr
37	Turn	High	11.78970, 78.02611	15 KM/Hr
38	Turn	High	11.78957, 78.02669	15 KM/Hr
39	Turn	High	11.78987, 78.02684	15 KM/Hr
40	Blind Spot	Blind Spot	11.79002, 78.03301	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	Government Hospital	11.7581107, 78.0180657	30 km/h	Medium
3	hospital	Government Hospital, Semmandapatty	11.79257, 78.029599	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45358, 77.85781



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45353, 77.85697



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45853, 77.85139



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46307, 77.84941



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46316, 77.84921



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45542, 77.81725



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.45631, 77.81668



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49224, 77.89606



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49153, 77.89615



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.48364, 77.88780



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49155, 77.88776



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49441, 77.88495



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49469, 77.88488



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.55614, 77.89764



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.58317, 77.90647



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63840, 77.94820



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64217, 77.95200



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67858, 77.96834



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.68925, 77.97742



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.68927, 77.97748



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.70218, 77.97866



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.70227, 77.97864



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.72565, 78.01770



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.73785, 78.03551



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.76193, 78.01473



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.78957, 78.02669



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.78987, 78.02684



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.79002, 78.03301

